Example: 12.20

A manufacturer estimates that, when units of a commodity are produced each month the total costs will be TC(Q) = 128 + 60Q + 8Q² Find the marginal cost, average cost, fixed cost, variable cost, average fixed cost and average variable cost.

Solution:

Given that TC(Q) = 128 + 60Q + 8Q²

We know TC = Fixed cost + variable cost

$$MC\ (Q) = \frac{d(TC)}{dQ}$$

$$= 0 + 60(1)Q^{1-1} + 8(2)Q^{2-1}$$

$$= 0 + 60Q^{0} + 16Q^{1} (\text{Since, } Q^{0}=1)$$

$$\boxed{MC = 60 + 16Q}$$

$$Average\ Cost = \frac{TC}{Q}$$

$$= \frac{128 + 60Q + 8Q^2}{Q}$$

$$AC = \frac{128}{Q} + 60 + 8Q$$

Constant value is known as fixed cost

Fixed cost = 128

FC = 128

$$\text{Average Fixed cost} = \frac{128}{Q}$$

$$AFC = \frac{128}{Q}$$

Average Variable cost =60 + 8Q (total variable cost divided by Q)

$$\therefore AVC = 60 + 8Q$$

12.4.8 Elasticity of Demand

Elasticity of Demand is the ratio of the proportionate change in quantity demanded to the proportionate change in price. In mathematical terms,

$$e_d = \left(\frac{P}{x}\right)\ \left(\frac{dx}{dp}\right)$$

In demand function Q = a–bP

$$e_d = (dQ/dP)(P/Q)$$

Example 12.21

If the demand function is $x = \frac{100}{P}$, find e_d with respect to price at the point where P = 2

Note

By taking supply function, the elasticity of supply can be calculated

Solution:

Given

$$x = \frac{100}{P} = 100P^{-1}$$

$$\therefore \frac{dx}{dp} = 100(-1)P^{-1-1}$$

$$= 100(-1)P^{-2}$$

$$= -100(P^{-2})$$

$$= \frac{-100}{P^2}$$

At P=2,

$$\frac{dx}{dp} = \frac{-100}{4} = -25 \quad \text{and} \quad x = \frac{100}{2} = 50$$

Substituting the values in formula

$$e_d = \frac{Pdx}{xdp} = \left(\frac{2}{50}\right)\left(\frac{-100}{4}\right) = \frac{-200}{200} = -1$$

$$e_d = -1$$

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